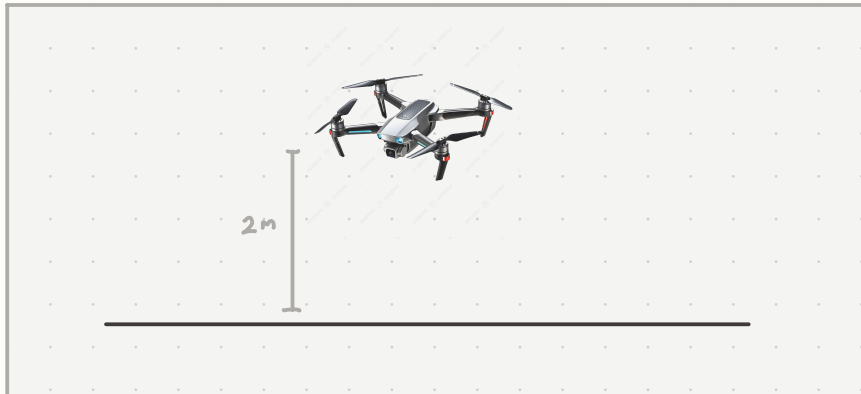


Exercise 1

Goal Today: Intro to Control Systems

Goal: Drone 2m above ground

- all day
- any weather



impossible?

② How does a drone fly? =>



- We describe a system by its input u and output y .
- A plant is a physical / mathematical system which turns an input into an output.
- The ideal case would be that $y = 2\text{m}$. This is called the reference r . Here, $r = 2\text{m}$.
- Gap between r and y is our error e .

$$e = r - y$$

- Point of CS: $e = 0$, $r = y$

GOAL

$$e = 0$$

① naive fix:

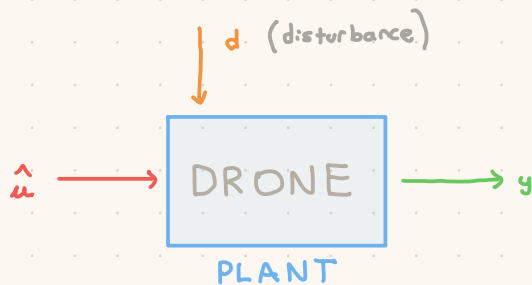
we calculate exact $\hat{u}$ that makes $y = 2m$, and we walk away.



when does this work? :

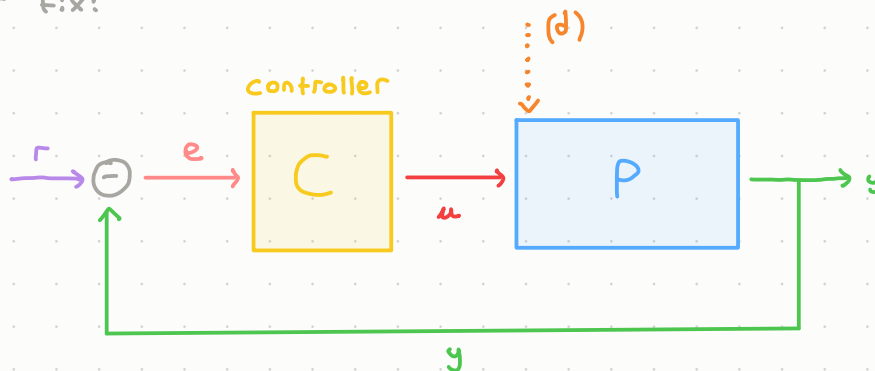
-
-

⚠ the villain: our environment



ex: gust of wind
pigeon landing
battery sagging } d
or
modeling errors

② clever fix:



- We can fully design the controller however we want. It can multiply, integrate, take the derivative, etc. of e and create a u .
- This is now a **closed loop** system. We call this action **Feedback**.
- Now, $u = F(y)$. We can **react**. **Disturbances** are rejected.

⚠ A CL architecture doesn't always solve the problem. The controller should be designed carefully or it may cause the system to become **unstable**.

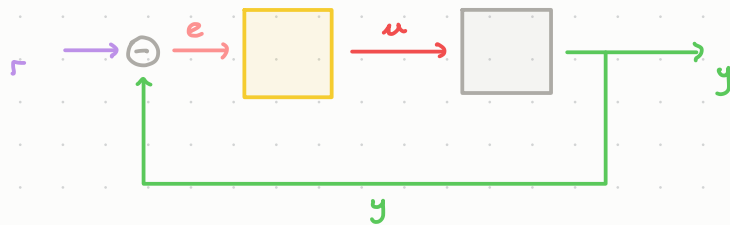
○ What have we learned? :

When dealing with **Control Systems**,

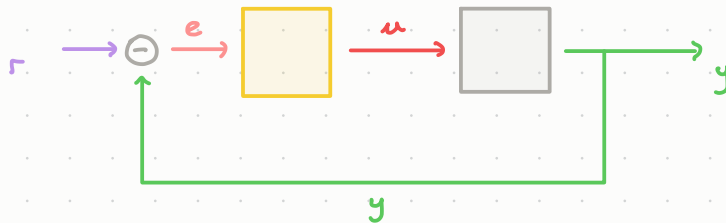
- What do I want?
- Can I measure what's happening?
 - OL.
 - C.L.
- What's fighting me?
- What can I change?

○ Let's apply these to new concepts:

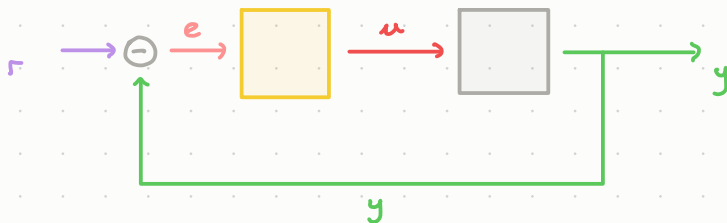
○ Cruise control



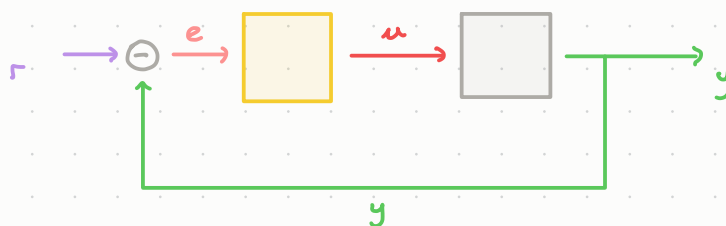
○ Shower



○ Toaster



○ Body Temp.



○ We have 3 objectives when designing a CS:

○ **Stabilization:** Make sure the plant does not deviate from desired behaviour.

○ **Performance:** Execute the task accurately.

○ **Robustness:** Perform well in presence of disturbances, modeling errors and noise.